

Assignment 3 Query Examples and Analysis

Queries that performed well:

1. “cristina lopes”
2. “machine learning”
3. “ACM”
4. “Master of software engineering”
5. “Inf 141”
6. “ICS Homepage”
7. “capstone projects”
8. “anteater express”
9. “artificial intelligence”
10. “information retrieval”

Queries that didn't perform well:

11. “research areas uci” (860.3 ms) → (47.8ms)

This query was slow because “research” and “uci” appear on many pages, so the search engine processed long posting lists. The ranking was weak because we didn't have good proximity scoring, so some random pages mentioning both words outranked the true overview pages as well.

Our Improvement: We added stopword removal, rarity-based postings intersection, and better proximity scoring to reduce the candidate set and improve ranking.

12. “software of engineering” (1413.7 ms) → (23.1ms)

This query was slow because all three terms have large posting lists, making intersections expensive. The ranking was poor because the real phrase is “software engineering,” and our model didn't recognize the phrase structure.

Our Improvement: We removed stopwords like “of” and improved bigram handling so the engine treats “software engineering” more meaningfully.

13. “information computer science” (2313.9 ms) → (169.6ms)

These were common words in the ICS corpus, so the query likely had to process many pages. Ranking suffered because we didn't understand the importance of missing connectors like “and,” so generic pages got ranked above department pages.

Our Improvement: Stopword removal and rare-term-first intersections shrink the candidate set, and improved proximity scoring helps group related terms together.

14. “science computer engineering” (582.4ms) → (42.2ms)

The terms were also extremely common, which likely caused slow intersections. The ranking was poor because the query mixed two different majors, and we lacked strong phrase/proximity detection.

Our Improvement: We improved phrase handling using bigrams and added proximity scoring to separate actual degree pages from noise.

15. “introduction programming” (630.3ms) → (7.7ms)

Both words are very common in course materials, so the posting lists are huge. Ranking was weak because we didn't reward the exact phrase “Introduction to Programming,” so the results were random, course-related pages.

Our Improvement: We added bigram boosting and better proximity checks, so true phrase matches rose to the top.

16. “uci of the engineering of the school” (7311.0 ms) → (28.4ms)

This was extremely slow because almost every word is high-frequency, especially “of,” leading to worst-case scanning, and the phrase was nonsensical. Ranking is noisy because the query contained mostly generic terms with no clear intent.

Our Improvement: Stopword removal eliminates “of/the,” and rare-term-first ranking dramatically reduces the number of documents we need to examine.

17. “uci research” (1107.3 ms) → (46.0ms)

This query was slow because both terms occurred on many pages, especially on faculty sites. Ranking was poor because our engine treated all pages with “research” equally, even if they weren't real research hubs.

Our Improvement: Using PageRank + term rarity reduced noise, and proximity scoring helped push actual research pages higher.

18. “University of California” (2386.2 ms) → (34.9ms)

These boilerplate words appeared in headers and footers of almost every page, so the posting lists are huge. Ranking is weak because the query is too generic, causing nearly every document to match.

Our Improvement: We drop stopwords and use rare-term-first intersection, making the query much faster and improving ranking stability.

19. “uc irvine” (365.9ms) → (16.9ms)

Both terms appear across the entire site, so the postings are large and slow to process. Ranking suffers because almost every page contains “UC Irvine,” making it hard to distinguish importance.

Our Improvement: Rare-term prioritization and lighter candidate sets help speed up processing and reduce random matches.

20. “Professor Lopes Christina research” (317.5ms) → (22.3ms)

It’s slow because “professor” and “research” are common words, and name components require scanning less common terms. Ranking was bad because the correct name is “Cristina Lopes,” and our engine didn’t handle name order or proximity.

Our Improvement: We improved proximity scoring and boosted rare terms first, so pages where the name appears correctly now rank higher.